

To prove part (c), we must prove the existence of a satisfying iteration $r \in [t+1]$. Let $r' \in [t]$ be the iteration that satisfies part (c) of the inductive hypothesis. If $r' = t$ and $x \notin P_{t+1}^-$ then we can let $r = t$. If $r' = t$ and $x \in P_{t+1}^-$ then we can let $r = t+1$. If $r' < t$ then we only have to show $x \notin P_{t+1}^-$, because in that case we can let $r = r'$. Since $r' < t$ we have $x \notin P_t^-$, and by part (a) we have $\bar{q}_{t-1}(x) > 0$. By Lemma 6 we have $\bar{q}_t(x) > 0$, and thus by part (a) again we have $x \notin P_{t+1}^-$. This concludes the proof of part (c).

To prove part (d), let $r \in [t+1]$ be the satisfying iteration from part (c). Note that $r < t+1$ by the premise of part (d). We first prove that

$$P_s^-(x)q_s(x) = 0 \text{ for all } s \in [r-1]. \quad (2)$$

Suppose for contradiction that Eq. (2) is not true, which implies that $P_s^-(x)q_s(x) > 0$ for some $s \in [r-1]$. By Lemma 6 and the fact that $\alpha > 0$ we have $\bar{q}_s(x) > 0$, which implies by part (a) that $x \notin P_{s+1}^-$, which contradicts part (c). Thus we have proved Eq. (2). We next prove that

$$P_r^-(x)q_r(x) = 1. \quad (3)$$

Suppose for contradiction that Eq. (3) is not true, which implies by part (c) and Lemma 8 that $P_r^-(x)q_r(x) = 0$. Thus by Eq. (2) we have $P_s^-(x)q_s(x) = 0$ for $s \in [r]$. We also have by part (c) that $P_s^+(x) = 0$ for $s \in [r]$. Thus by Lemma 6 we have $\bar{q}_r(x) = 0$, and this implies by part (a) that $x \in P_{r+1}^-$, which by $r < t+1$ contradicts part (c). Thus we have proved Eq. (3). We are now ready to complete the proof of part (d). We have

$$\begin{aligned} \bar{q}_{t+1}(x) &= \frac{\sum_{s=1}^{t+1} \lambda_s P_s^-(x)q_s(x) + P_s^+(x)q_s^+(x)}{\sum_{s=1}^{t+1} \lambda_s P_s^-(x) + P_s^+(x)} && \text{Lemma 6} \\ &= \frac{\sum_{s=1}^{t+1} \frac{\alpha}{|D_s|} P_s^-(x)q_s(x) + P_s^+(x)q_s^+(x)}{\sum_{s=1}^{t+1} \frac{\alpha}{|D_s|} P_s^-(x) + P_s^+(x)} \\ &= \frac{\frac{\alpha}{|D_r|} + \sum_{t=r+1}^{t+1} 1}{\sum_{s=1}^r \frac{\alpha}{|D_s|} + \sum_{t=r+1}^{t+1} 1} && \text{Part (c), Lemma 8, Eq. (2) and Eq. (3)} \\ &= \frac{\frac{\alpha}{|D_r|} + t - r + 1}{\sum_{s=1}^r \frac{\alpha}{|D_s|} + t - r + 1} \\ &= \frac{\frac{\alpha}{|D_r|} + t - r + 1}{\sum_{s=1}^r \frac{\alpha}{|D_s|} + t - r + 1} \\ &= \frac{\frac{\alpha}{|P_r^-|} + t - r + 1}{\sum_{s=1}^r \frac{\alpha}{|P_s^-|} + t - r + 1} && \text{Definitions of } P_t^- \text{ and } P_t^+ \\ &\geq \frac{\frac{\alpha}{|P_r^-|} + t - r + 1}{\frac{\alpha}{|P_r^-|} \sum_{s=1}^r (1 - \beta)^{r-s} + t - r + 1} && \text{Part (b)} \\ &= \frac{\frac{\alpha}{|P_r^-|} + t - r + 1}{\frac{\alpha}{|P_r^-|} \sum_{s=0}^{r-1} (1 - \beta)^s + t - r + 1} \\ &= \frac{\frac{\alpha}{|P_r^-|} + t - r + 1}{\frac{\alpha(1 - (1 - \beta)^r)}{\beta |P_r^-|} + t - r + 1} && \text{Geometric series formula} \end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha + |P_r^-|(t - r + 1)}{\frac{\alpha(1-(1-\beta)r)}{\beta} + |P_r^-|(t - r + 1)} \\
&\geq \frac{\alpha + t - r + 1}{\frac{\alpha(1-(1-\beta)r)}{\beta} + t - r + 1} \quad \text{Lemma 9 and } |P_r^-| \geq 1 \text{ (by choice of } r)
\end{aligned}$$

which proves the first inequality of part (d). Continuing from above

$$\begin{aligned}
\bar{q}_{t+1}(x) &\geq \frac{\alpha + t - r + 1}{\frac{\alpha(1-(1-\beta)r)}{\beta} + t - r + 1} && \text{From above} \\
&\geq \frac{\alpha}{\frac{\alpha(1-(1-\beta)r)}{\beta}} && \text{Lemma 9} \\
&= \frac{\beta}{1 - (1 - \beta)^r} && \alpha > 0 \\
&\geq \beta && \beta > 0
\end{aligned}$$

which proves the second inequality of part (d). $\square$

We are now ready to complete the proof of Theorem 5. Assume that event E occurs, which by Lemma 10 happens with probability at least $1 - \frac{1}{T}$. For each prompt $x \in P$ let r_x be the iteration that satisfies Lemma 11(c) when the lemma is applied to prompt x and iteration T . Let $r = \frac{\log(2/\varepsilon)}{\beta}$, and note that by assumption $r < T$. We have

$$\begin{aligned}
&\Pr_{x \sim P, y \sim g_T(x)}[q(x, y) = 1] \\
&= \mathbb{E}_{x \sim P, y \sim g_T(x)}[q(x, y)] \\
&= \mathbb{E}_{x \sim P} \left[\sum_y \mathcal{D}_T(y|x) q(x, y) \right] && \text{Definition 1} \\
&= \mathbb{E}_{x \sim P}[\bar{q}_T(x)] \\
&\geq \mathbb{E}_{x \sim P}[\bar{q}_T(x) \mid r_x \leq r] \Pr_{x \sim P}[r_x \leq r] \\
&= \mathbb{E}_{x \sim P}[\bar{q}_T(x) \mid r_x \leq r] \Pr_{x \sim P}[x \notin P_{r+1}^-] && \text{Lemma 11(c)} \\
&= \mathbb{E}_{x \sim P}[\bar{q}_T(x) \mid r_x \leq r] \left(1 - \frac{|P_{r+1}^-|}{|P|} \right) \\
&= \mathbb{E}_{x \sim P}[\bar{q}_T(x) \mid r_x \leq r] \left(1 - \frac{|P_{r+1}^-|}{|P_1^-|} \right) && \text{Lemma 11(a)} \\
&\geq \mathbb{E}_{x \sim P}[\bar{q}_T(x) \mid r_x \leq r] (1 - (1 - \beta)^r) && \text{Lemma 11(b)} \\
&\geq \min_{x: r_x \leq r} \frac{\alpha + T - r_x}{\frac{\alpha(1-(1-\beta)r_x)}{\beta} + T - r_x} (1 - (1 - \beta)^r) && \text{Lemma 11(d)} \\
&\geq \min_{x: r_x \leq r} \frac{\alpha + T - r_x}{\frac{\alpha}{\beta} + T - r_x} (1 - (1 - \beta)^r) \\
&\geq \frac{\alpha + T - r}{\frac{\alpha}{\beta} + T - r} (1 - (1 - \beta)^r) && \text{Lemma 9}
\end{aligned}$$

$$\begin{aligned}
&\geq \frac{\alpha + T - r}{\frac{\alpha}{\beta} + T - r} \left(1 - e^{-\beta r}\right) \\
&= \frac{\alpha + T - r}{\frac{\alpha}{\beta} + T - r} \left(1 - \frac{\varepsilon}{2}\right)
\end{aligned}$$

Since

$$T \geq \frac{\log(2/\varepsilon)}{\beta} + \frac{2\alpha}{\beta\varepsilon} = r + \frac{2\alpha}{\beta\varepsilon}$$

it is easy to show via algebra that

$$\frac{\alpha + T - r}{\frac{\alpha}{\beta} + T - r} \geq 1 - \frac{\varepsilon}{2}$$

and plugging this into the final expression above proves $\Pr_{x \sim P, y \sim g_T(x)}[q(x, y) = 1] \geq \left(1 - \frac{\varepsilon}{2}\right)^2 \geq 1 - \varepsilon$, which proves the theorem.

B Additional Plots

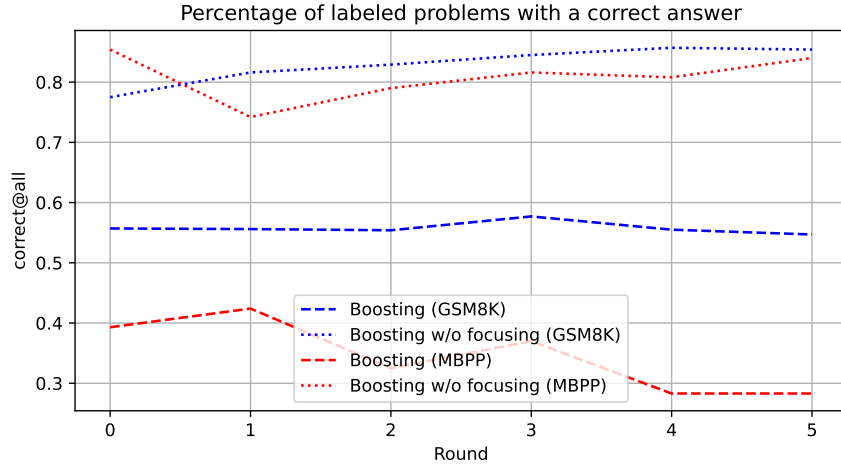


Figure 3: Labeler accuracy across rounds. These results use *weak data* (A). Since training accuracy increases across rounds, the weak labeler gets more queries per question in both cases. Despite this, for *Boosting* we see that accuracy is relatively constant for GSM8K and decreasing for MBPP. This is because we focus on increasingly harder problems. In *Boosting w/o focusing*, we observe labeler accuracy increasing because we do not focus labeler efforts on the highest difficulty problems.

In Figure 4, we plot average response length on the test set as a function of training rounds. Average response length does not change significantly throughout training, and incorrect answers are longer on average.